

Reading the Storm: Media Attribution of U.S. Climate Disasters, 2000–2024

Hari Gunda¹, Michael Price², Angela Doku³, and John List⁴

¹TODO: Affiliation

²University of Alabama

³University of Toronto

⁴University of Chicago

June 13, 2026

Abstract

TODO (150–250 words). Motivation: how local and national newspapers frame the causes of billion-dollar U.S. climate disasters – anthropogenic vs. natural/random attribution – and what drives variation in that framing. Briefly state data (NOAA Billion-Dollar Disasters, 2000–2024, matched to newspaper coverage via Google Custom Search), method (NLP-based Attribution Index at the article and disaster level), and headline findings (to be filled in).

Keywords: climate change attribution, media framing, natural language processing, disaster economics

1 Introduction

TODO: Motivate the question — do U.S. newspapers attribute billion-dollar weather/climate disasters to anthropogenic climate change, or frame them as natural/random variation? Why does this matter (public risk perception, policy salience, insurance markets, disaster preparedness)? Preview data, method, and main results. Outline the rest of the paper.

2 Related Literature

TODO: Position relative to (i) climate communication / media framing literature, (ii) economics of natural disasters and salience, (iii) text-as-data / NLP methods in economics, (iv) political economy of climate policy and local media markets.

3 Data

3.1 NOAA Billion-Dollar Disasters Dataset

The sample is drawn from NOAA’s Weather and Climate Billion-Dollar Disasters dataset, restricted to U.S. events occurring between 2000 and 2024 (313 events). Each event is characterized by a disaster type (Severe Storm, Tropical Cyclone, Flooding, Drought, Wildfire, Winter Storm, or Freeze), a begin and end date, the set of affected states, CPI-adjusted and unadjusted cost (in millions of 2024 dollars), and a fatality count. Severe storms are the most common event type (179 of 313 events), followed by tropical cyclones (48), flooding (30), drought (21), wildfire (19), winter storm (13), and freeze (3). Total CPI-adjusted damages across the sample exceed \$2.36 trillion, with a median event cost of \$2.4 billion (mean \$7.5 billion, driven by a right tail of extreme events up to \$201.3 billion); total reported fatalities across all events are 10,849.

3.2 Newspaper Coverage Data

Newspaper universe. The candidate universe of outlets is drawn from the Access World News (AWN) database, which lists 3,647 U.S. newspapers and wire services with associated city, state, and language metadata. Of these, 970 had no listed website and were excluded. The remaining outlets were passed through an automated verification pipeline that attempted to resolve and HTTP-check each listed URL (classifying each as verified, verified via HTTP-only, or unverified with a status code/reason); 899 outlets ultimately produced a usable, live domain.

Programmable Search Engine (PSE) construction. The 899 verified domains were partitioned into 18 regional Google Programmable Search Engines (PSEs), grouped by the states each outlet primarily serves, plus a 19th “National” PSE covering major national and wire outlets (e.g., the *New York Times*, *Washington Post*, CNN, AP, NPR, USA Today) that are relevant regardless of event geography. Each of the 313 disaster events is mapped to the subset of regional PSEs corresponding to its affected states (plus the National PSE), so that searches are restricted to outlets plausibly covering that event’s geography.

Collection pipeline. For each event, queries combining the disaster name/type and date window are issued against the relevant PSE subset via the Google Custom Search JSON API. Candidate results are filtered on (i) a date window around the event’s begin/end dates (with retrospective coverage allowed up to the start of the next event of the same type, where applicable), (ii) geographic relevance – results must reference a place name consistent with the event’s affected states or broader region, and (iii) disaster-type/keyword relevance. Up to 30 candidate articles are retained per event. The full collection run returned 1,694 candidate articles across the 313 events: 237 events (75.7%) returned at least one candidate article (mean 5.4 articles per event among events with any coverage; 76 events returned none). Of the 1,694 candidate articles, 911 (53.8%) are from outlets on the National PSE list and 783 (46.2%) are from regional/local outlets. These candidate articles are then passed through the relevance-screening and attribution-coding procedure described in Section 4.

3.3 Constructing the Attribution Index

Each of the 1,694 candidate articles is first screened for *relevance* – whether it substantively discusses the specific NOAA-listed event it was retrieved for (correct disaster, geography, and time window), versus being off-topic, ambiguous (**Borderline**), or unusable (**Trash**). Only articles coded **Relevant** proceed to attribution coding.

Each relevant article is then assigned to one of five categories describing how it frames the causal origin of the event, with an associated numeric score $a_{i,d} \in \{-1, 0, 0.5, 1\}$:

Article-level scores are averaged to a disaster-level *Attribution Index*; the aggregation formula, calibration procedure, and treatment of events with no relevant coverage are described in Section 4.

3.4 Summary Statistics

Table 2 summarizes the collection-stage data described above. The Attribution Index distribution itself (mean, SD, and quantiles of $\text{AttributionIndex}_d$) will be added once the relevance-

Table 1: Attribution Coding Categories

Code	Description	Score
A	Explicit anthropogenic climate-change attribution	+1.0
B	Hedged / contextual climate mention	+0.5
C	No causal attribution (pure event reporting)	0.0
D	Explicit non-climate attribution (natural variability or other human causes)	−1.0
E	Mixed / contested framing	0.0 (flagged)

Notes: Full category definitions, decision rules, and example language are given in the project codebook ([paper/attribution_codebook.md](#)). Category E receives the same numeric score as C but is separately flagged as *contested* when computing disaster-level dispersion.

screening and attribution-coding pass described in Section 4 is complete.

Table 2: Summary Statistics: Sample and Coverage

	Value
NOAA billion-dollar disaster events (2000–2024)	313
Candidate newspaper articles collected	1,694
Events with ≥ 1 candidate article	237 (75.7%)
Events with no candidate articles	76 (24.3%)
Mean articles per covered event	5.4
Maximum articles for a single event	30
Articles from National PSE outlets	911 (53.8%)
Articles from regional/local PSE outlets	783 (46.2%)
<i>Pending attribution-coding pass:</i>	
Mean Attribution Index, $\text{AttributionIndex}_d$	TBD
Share of disasters with Contested coverage	TBD

Notes: Coverage statistics describe the raw candidate-article sample prior to relevance screening (Section 4).

4 Empirical Approach

4.1 Article-Level Classification

Each candidate article is coded in two sequential steps, following a written codebook (reproduced in full in the project repository as [attribution_codebook.md](#)) that is also used as the basis for the LLM coding prompt.

Step 1: Relevance screening. Every candidate article is classified as **Relevant** (substantively discusses the specific NOAA-listed event – correct disaster, geography, and time window), **Borderline** (plausibly related but ambiguous, e.g. a brief wire item or a retrospective mention), or **Trash** (off-topic, wrong event, duplicate, paywall/landing page, or dead link). Only **Relevant** articles proceed to Step 2; **Borderline** and **Trash** articles are retained in the data for transparency but excluded from the index.

Step 2: Attribution coding. Each **Relevant** article is assigned to one of five categories describing how it frames the causal origin of the event (Table 1): (A) explicit anthropogenic attribution (+1.0), (B) hedged or contextual climate mention (+0.5), (C) no causal attribution (0.0), (D) explicit non-climate attribution to natural variability or other human causes (−1.0), and (E) mixed/contested framing (0.0, separately flagged as **Contested**). For category D, coders additionally record whether the non-climate attribution is to natural variability (“just weather”) or to other human causes (e.g. land use, infrastructure), since both score −1 on the climate-vs-natural axis but imply different policy narratives.

Calibration and validation. Coding is performed by an LLM following the codebook, with a staged validation procedure that applies two different reliability bars: a low gate during calibration to catch codebook problems cheaply, and a higher target for the reliability statistics reported in the paper. First, the researcher hand-codes a stratified random calibration sample of approximately 50 articles (stratified by disaster type and by pre-/post-2010 period) for both Relevance and Attribution category. Second, the LLM codes the same 50 articles from the codebook alone. Third, human and LLM codes are compared using unweighted Cohen’s κ for the categorical Relevance (3-category) and Attribution (5-category) codes; if agreement is weak ($\kappa < 0.6$, “moderate” or worse on the Landis and Koch (1977) scale), the codebook wording and examples are revised and the comparison is repeated before scaling up. This 0.6 threshold is intentionally low – its purpose is to catch broken category definitions early, not to certify reliability. Once this gate is cleared, the LLM codes the remaining articles in the full sample. Finally, an independent post-hoc audit sample (approximately 100 articles, drawn after the full pass) is hand-coded and compared to the LLM codes. For this audit sample we report unweighted Cohen’s κ for Relevance (target $\kappa \geq 0.7$); for Attribution, we report both unweighted κ and a weighted κ (linear or quadratic weights) that reflects the ordinal structure of the underlying scores $\{-1, 0, 0.5, 1\}$, since unweighted κ treats an A-vs-D disagreement the same as a B-vs-C disagreement despite the very different implied score gap (target $\kappa_w \geq 0.7$, ideally ≥ 0.8); and we additionally report the mean absolute difference between human- and LLM-assigned numeric scores as a directly inter-

pretable companion statistic. Should the audit-sample statistics fall short of these targets, this is disclosed and discussed as a limitation, since the boundaries between categories B/C and C/E are genuinely fuzzy even for human coders.

4.2 Disaster-Level Attribution Index

For each disaster d with N_d articles coded **Relevant**, article-level scores are averaged to a disaster-level *Attribution Index*:

$$\text{AttributionIndex}_d = \frac{1}{N_d} \sum_{i=1}^{N_d} a_{i,d}, \quad a_{i,d} \in \{-1, 0, 0.5, 1\} \quad (1)$$

where $a_{i,d}$ is the numeric score for article i covering disaster d from Table 1, with -1 indicating explicit non-climate (natural/random or other human-cause) attribution and $+1$ indicating explicit anthropogenic climate-change attribution. Category E (mixed/contested) scores 0 , identically to category C (no attribution), but is additionally flagged so that a disaster-level *Contested share* – the fraction of N_d articles coded E – can be reported separately as a measure of framing disagreement, distinct from the simple absence of any causal claim.

For the 76 events with $N_d = 0$ (no relevant articles, including events that returned zero search results entirely), $\text{AttributionIndex}_d$ is left undefined (missing) rather than imputed, and these events are analyzed as a separate “no media attribution” group rather than dropped from the sample. All of the above quantities – N_d , $\text{AttributionIndex}_d$, and the Contested share – are computed via formulas in the **All Disasters** sheet of the project workbook (`attribution.xlsx`), which references the article-level codes in the **Master** sheet.

4.3 Specification

TODO: Main regression specification relating the Attribution Index to disaster characteristics (type, severity/cost, geography, time), e.g.

$$\text{AttributionIndex}_d = \beta_0 + \beta_1 \text{Cost}_d + \beta_2 \text{Type}_d + \beta_3 \text{Year}_d + \mathbf{X}_d' \boldsymbol{\gamma} + \varepsilon_d \quad (2)$$

5 Results

5.1 Descriptive Patterns

TODO: Figure 1 – Attribution Index over time, 2000–2024.

Figure 1: Attribution Index Over Time

5.2 Variation by Geography

TODO: Coastal vs. inland, red vs. blue states.

5.3 Variation by Disaster Type and Severity

TODO: Wildfires/hurricanes vs. floods/severe storms; cost as a predictor of anthropogenic framing.

5.4 Main Regression Results

TODO: Table 3.

Table 3: Determinants of the Attribution Index

	(1)	(2)	(3)
TODO			
Notes: TODO. Standard errors in parentheses.			

6 Robustness

TODO: Alternative classifiers, sample restrictions (national vs. local outlets only), placebo checks.

7 Discussion

TODO: Interpret findings in light of climate communication, risk perception, and policy implications discussed in the Introduction.

8 Conclusion

TODO: Summarize contribution, limitations, and directions for future work.

References

Landis, J. R. and Koch, G. G. (1977). The Measurement of Observer Agreement for Categorical Data. *Biometrics*, 33(1):159–174.

A Data Appendix

A.1 Disaster Event List

TODO: Full list of 313 NOAA events with assigned states/regions and PSE engine coverage.

A.2 Newspaper Source List

TODO: Description of the 19 Programmable Search Engines and their domain coverage.

A.3 Additional Tables and Figures

TODO.